DESIGN 06 OF 10

CARBON MATRIX

A continuous 2/2 twill weave with selective reinforcement patches.

THE DESIGN

A woven-carbon look built as real geometry: every tow rises over and dives under its neighbours in a 2/2 twill. Smooth, slightly raised patches suggest reinforcement where the palm loads the handle.



CLOSE-UP · MID BODY

61.2 g
PLA, SEALED CORE

60
TOWS

2.04 mm
TOW PITCH



THE HANDLE

A sleeve that slides over your paddle's flat handle: a 30.2 × 16 mm slot (the 30 mm wide handle plus 0.2 mm, 16 mm thick), 130 mm deep so the paddle seats fully, with a solid 5 mm base and a counterbored 3.5 mm hole that takes a low-head M3 cap screw flush. The head is solid, with both short side walls of the slot slit open at the top so the flared throat of the paddle slides in smoothly; a sealed hollow core lightens the rest. The outer shape is the same on all ten designs.

135 mm	30 × 16 mm	130 mm	M3 low-head
LENGTH	PADDLE SLOT	SLOT DEPTH	SCREW, FLUSH

CHOOSE YOUR COLOUR

DESIGNER'S PICK: JET BLACK



Carbon Matrix, all round.

Four views, four surfaces and the idea behind the design. Shown as it prints: one solid PLA colour.



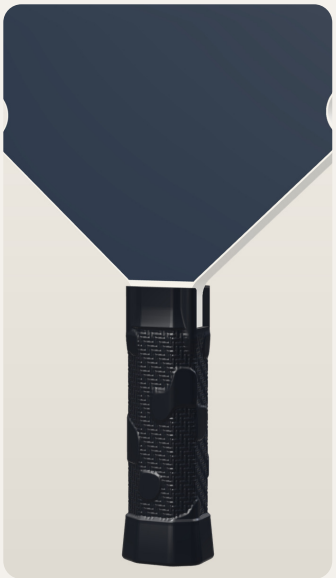
3/4 VIEW · TURN 0°



SIDE VIEW · TURN 90°



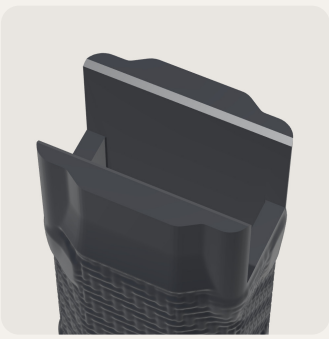
CUTAWAY · HOLLOW CORE



ON A PADDLE · ILLUSTRATION



CLOSE-UP · MID BODY



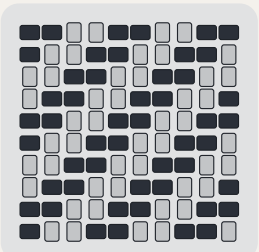
RIM & SLOT · FROM ABOVE



UNROLLED · LIT RELIEF



BUTT · Ø3.5 HOLE + CBORE



DESIGN INSPIRATION

Carbon-fibre cloth, the material many paddle faces are made from. The twill is printed rather than woven, so it feels like a fine fabric under the hand.

WHY IT GRIPS

- 60 tows around at a 2.04 mm pitch, 2/2 twill
- Weave relief about 0.4 mm, the finest texture in the set
- Smooth patches stand about 0.5 mm above the weave

PRINT TIP

The weave is the finest texture in the set: 0.4 mm of relief on a 2 mm pitch.

THE WHOLE COLLECTION



The paddle in the illustration is generic, drawn with your 30 × 16 mm handle. Every design prints in any of the ten PLA colours shown on page 1. 100 % infill · 0.20 mm layers · upright, butt end down.